

REPRESENTATION-LEVEL UNIQUENESS FOR $\text{Cay}(S_n, C(n, k; r))$: A PROOF OF A CONJECTURE OF LI, XIA AND ZHOU

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ABSTRACT. For $1 \leq r < k < n$ let $C(n, k; r)$ be the set of k -cycles of S_n whose support contains $\{1, \dots, r\}$. Li, Xia and Zhou proved that, for $4 \leq k \leq n - 2$, the strictly second largest eigenvalue of the Cayley graph $\text{Cay}(S_n, C(n, k; r))$ equals the value $\mu_2(n, k; r)$ carried by the standard representation, and conjectured (their Conjecture 4.7) that for all $n \geq 5$ and $1 \leq r < k \leq n - 2$ this value is attained *only* by the standard representation when k is even, and exactly by the standard representation and its sign twist when k is odd. We prove this conjecture. The missing ingredient in the published induction is strictness at the base case $r = k - 1$; we obtain it for every $k \geq 4$ and every $n \geq k + 1$ by an equality analysis in Weyl's inequality, after which the published induction on $n - r$ becomes strict. The cases $k = 2, 3$ are handled directly with Jucys–Murphy elements, and two small groups ($n = 6, 7$) are settled by an exact computation that reproduces the published machine check.

1. INTRODUCTION

Let $1 \leq r < k \leq n$ and let

$$C(n, k; r) = \{\sigma \in S_n : \sigma \text{ is a } k\text{-cycle and } \{1, \dots, r\} \subseteq \text{supp}(\sigma)\}.$$

This inverse-closed set was introduced by Siemons and Zalesski [4], who computed the spectrum of $C(n, k; r)^+ = \sum_{h \in C(n, k; r)} h$ on the natural permutation module. The largest eigenvalue of $C(n, k; r)^+$ on the standard representation is

$$(1) \quad \mu_2(n, k; r) = (k - 2)! \binom{n - r}{k - r} \frac{1}{n - r} \left((k - 1)(n - k) - \frac{(k - r - 1)(k - r)}{n - r - 1} \right)$$

([4]; see [3, (9) and the remark after Lemma 2.10]). A Cayley graph on S_n has the *Aldous property* if its strictly second largest eigenvalue (the largest eigenvalue strictly below the degree) is attained by the standard representation. Li, Xia and Zhou [3, Theorem 1.3] proved that $\text{Cay}(S_n, C(n, k; r))$ has the Aldous property for $n \geq 5$, $k \geq 4$, $1 \leq r < k \leq n - 1$, with the exceptions $(n, k, r) = (6, 5, 1)$ and $(n \text{ odd}, k = n - 1, r < n/2)$. They proposed the following sharper statement.

Conjecture 1.1 ([3, Conjecture 4.7]). *Suppose that $n \geq 5$ and $1 \leq r < k \leq n - 2$. When k is even, the second largest eigenvalue of $\text{Cay}(S_n, C(n, k; r))$ is attained uniquely by the standard representation $\rho_{(n-1,1)}$. When k is odd, the strictly second largest eigenvalue of $\text{Cay}(S_n, C(n, k; r))$ is attained exactly by $\rho_{(n-1,1)}$ and $\rho_{(2,1^{n-2})}$.*

As the authors explain after the conjecture, their inductive argument only yields the non-strict bound $\lambda \leq \mu_2(n, k; r)$ for the eigenvalues of the other irreducible blocks, and the conjecture requires strict inequality. They verified it for $k = n - 2$, n even and $1 \leq r \leq (n - 1)/2$ [3, Lemmas 4.2, 4.3].

Theorem 1.2. *Conjecture 1.1 is true.*

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Strategy. The published proof of the Aldous property is an induction on $n - r$ whose base is the case $r = k - 1$ [3, Theorem 1.2], and whose step splits $C(n, k; r)$ into the cycles fixing a point $t \notin [r]$ and the cycles moving it (Lemma 2.6). Because the base is only non-strict, so is every conclusion. We prove a *strict* base for every $k \geq 4$ and $n \geq k + 1$ (Proposition 4.3); strictness then propagates through the same split (Lemma 3.1). The strict base is itself proved by induction on n : the equality case of Weyl's inequality forces a common eigenvector, which lies in a standard child of $(n - 2, 2)$ or $(n - 2, 1, 1)$ and is invariant under the symmetric (or alternating) group on a k -subset; two explicit models show that no such vector exists (Lemmas 4.1 and 4.2). This argument generalises the one used for the single layer $n - k = 3$ in an earlier note produced with the EULER system. The layer $k = n - 2$ needs extra care because the published results for $k = n - 1$ have exceptions; there we combine the strict step with [3, Lemmas 4.2, 4.3] and a finite computation. Finally, $k = 2, 3$ are handled directly (Section 5).

2. SETUP AND PUBLISHED INPUTS

2.1. Blocks. For a finite group G and an inverse-closed $S \subseteq G \setminus \{1\}$, the adjacency operator of $\text{Cay}(G, S)$ (edges $\{g, gs\}$) is right multiplication by S^+ on $\mathbb{C}[G]$, and its spectrum is the union, over irreducible representations ρ , of $\dim \rho$ copies of the spectrum of $\rho(S^+)$ [3, Proposition 2.1]. An eigenvalue is *attained by* ρ if it is an eigenvalue of $\rho(S^+)$. We always use an orthogonal realisation of ρ , so $\rho(S^+)$ is real symmetric.

Fix $1 \leq r < k < n$ and put $H = C(n, k; r)$. For $\lambda \vdash n$ let

$$T_\lambda = T_\lambda(n, k, r) = \rho_\lambda(H^+), \quad \Lambda_\lambda = \Lambda_\lambda(n, k, r) = \lambda_{\max}(T_\lambda).$$

By (1) and the cited remark, $\Lambda_{(n-1,1)} = \mu_2(n, k; r)$. A k -cycle has sign $(-1)^{k-1}$. Since $\rho_{\lambda'} \cong \rho_\lambda \otimes \text{sgn}$, the block $T_{\lambda'}$ is orthogonally similar to T_λ if k is odd and to $-T_\lambda$ if k is even. By [3, Lemma 2.14], H generates S_n if k is even and A_n if k is odd, so the degree $|H|$ has multiplicity 1, resp. 2, in the graph, and it is attained only by (n) , resp. by (n) and (1^n) . Define

$$\mathcal{D}_k(n) = \begin{cases} \{(n)\} & k \text{ even,} \\ \{(n), (1^n)\} & k \text{ odd,} \end{cases} \quad \Theta_k(n) = \begin{cases} \{(n-1, 1)\} & k \text{ even,} \\ \{(n-1, 1), (2, 1^{n-2})\} & k \text{ odd.} \end{cases}$$

Every $\lambda \in \Theta_k(n)$ has $\Lambda_\lambda = \mu_2(n, k; r)$, and every eigenvalue of every block T_λ with $\lambda \notin \mathcal{D}_k(n)$ is strictly smaller than $|H|$.

Definition 2.1. For $1 \leq r < k < n$ we say that

- $\mathbf{N}(n, k, r)$ holds if $\Lambda_\lambda \leq \mu_2(n, k; r)$ for every $\lambda \vdash n$ with $\lambda \notin \mathcal{D}_k(n)$;
- $\mathbf{S}(n, k, r)$ holds if $\Lambda_\lambda < \mu_2(n, k; r)$ for every $\lambda \vdash n$ with $\lambda \notin \mathcal{D}_k(n) \cup \Theta_k(n)$.

Lemma 2.2. $\text{Cay}(S_n, C(n, k; r))$ has the Aldous property if and only if $\mathbf{N}(n, k, r)$ holds. The assertion of Conjecture 1.1 for (n, k, r) is equivalent to $\mathbf{S}(n, k, r)$, and $\mathbf{S}(n, k, r)$ implies $\mathbf{N}(n, k, r)$.

Proof. The eigenvalues strictly below $|H|$ are exactly the eigenvalues of the blocks T_λ , $\lambda \notin \mathcal{D}_k(n)$, together with the non-top eigenvalues of the degree blocks (there are none: degree blocks are one-dimensional). Hence the strictly second largest eigenvalue is $\max_{\lambda \notin \mathcal{D}_k(n)} \Lambda_\lambda \geq \Lambda_{(n-1,1)} = \mu_2$, which gives the first statement. If $\mathbf{S}$ holds, this maximum equals μ_2 and is attained exactly by $\Theta_k(n)$; when k is even the graph is connected, so the second largest and strictly second largest eigenvalues coincide. Conversely, if μ_2 is the strictly second largest eigenvalue and is attained only by $\Theta_k(n)$, every other non-degree block has all eigenvalues $\leq \mu_2$ and $\neq \mu_2$. $\square$

2.2. Tools.

Lemma 2.3 (Weyl's inequality with equality). *Let A, B be real symmetric operators on the same space. Then $\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B)$, with equality if and only if some nonzero v satisfies $Av = \lambda_{\max}(A)v$ and $Bv = \lambda_{\max}(B)v$.*

Proof. For a unit top eigenvector v of $A + B$, $\lambda_{\max}(A + B) = \langle Av, v \rangle + \langle Bv, v \rangle$ and each Rayleigh quotient is at most the corresponding top eigenvalue, with equality only for a top eigenvector. The converse is immediate. $\square$

Lemma 2.4 (Branching). *Let $N \geq 5$, $\lambda \vdash N$ and $t \in [N]$. The restriction of S^λ to $\text{Sym}([N] \setminus \{t\}) \cong S_{N-1}$ is multiplicity free, $S^\lambda \downarrow \cong \bigoplus_\nu S^\nu$ over the partitions ν obtained from λ by removing one box (children). In particular: the trivial module $S^{(N-1)}$ is a child only of $(N), (N-1, 1)$; the sign module $S^{(1^{N-1})}$ only of $(1^N), (2, 1^{N-2})$; the standard module $S^{(N-2, 1)}$ only of $(N-1, 1), (N-2, 2), (N-2, 1, 1)$; and $S^{(2, 1^{N-3})}$ only of $(2, 1^{N-2}), (2, 2, 1^{N-4}), (3, 1^{N-3})$.*

Lemma 2.5 (A full cycle class). *Let $Y \subseteq [N]$ with $m = |Y| \geq 4$, let K_Y be the set of all m -cycles of $\text{Sym}(Y)$, so $|K_Y| = (m-1)!$, and let ρ be an orthogonal representation of S_N . Then $\lambda_{\max}(\rho(K_Y^+)) \leq (m-1)!$, and v is an eigenvector for the value $(m-1)!$ if and only if v is fixed by $\text{Sym}(Y)$ (when m is even) or by $\text{Alt}(Y)$ (when m is odd).*

Proof. For a unit vector v , $\langle \rho(K_Y^+)v, v \rangle = \sum_{g \in K_Y} \langle \rho(g)v, v \rangle \leq |K_Y|$, with equality iff $\rho(g)v = v$ for all $g \in K_Y$. The group generated by K_Y is $\text{Sym}(Y)$ for m even and $\text{Alt}(Y)$ for m odd [3, Lemma 2.14]. Conversely a fixed vector is an eigenvector with eigenvalue $|K_Y|$. $\square$

Lemma 2.6 (The split). *Let $1 \leq r \leq k-1$, $k \leq n-1$, and $t = r+1$. Then $C(n, k; r) = C_t \sqcup C'_t$, where C_t consists of the elements fixing t and C'_t of those moving t . Under an order-preserving relabelling $[n] \setminus \{t\} \rightarrow [n-1]$ (which fixes $[r]$), C_t becomes $C(n-1, k; r)$. If $r \leq k-2$ then $C'_t = C(n, k; r+1)$; if $r = k-1$ then $C'_t = K_Y$ with $Y = [k]$. Consequently, for $\lambda \vdash n$, the operator $\rho_\lambda(C_t^+)$ is orthogonally similar to $\bigoplus_\nu T_\nu(n-1, k, r)$, the sum over the children ν of λ , and each summand is supported on the corresponding isotypic component of $S^\lambda \downarrow_{\text{Sym}([n] \setminus \{t\})}$.*

Proof. A k -cycle moving $[r]$ either fixes t or moves $[r] \cup \{t\} = [r+1]$. The last sentence follows from Lemma 2.4, since $C_t \subseteq \text{Sym}([n] \setminus \{t\})$. $\square$

2.3. Published inputs. We use the following results of [3].

Input 2.7 (Base values). $\mu_2(n, k; k-1) = (k-1)!(n-k)$ for $2 \leq k \leq n-1$ [3, Theorem 1.2]; this also follows from (1).

Input 2.8 (Recurrence [3, (10)]). $\mu_2(n, k; r) = \mu_2(n-1, k; r) + \mu_2(n, k; r+1)$ for $1 \leq r \leq k-2$ and $k \leq n-2$.

Input 2.9 (The layer $k = n-1$ [3, Theorems 3.3, 3.5]). Let $m \geq 5$ and $1 \leq r \leq m-2$, and consider $C(m, m-1; r)$.

- (a) If m is odd and $m/2 < r < m-1$, then the second largest eigenvalue is $r(m-3)! = \mu_2(m, m-1; r)$ and is attained uniquely by $(m-1, 1)$. Hence $\mathbf{S}(m, m-1, r)$ and $\mathbf{N}(m, m-1, r)$ hold.
- (b) If $m \geq 6$ is even and $(m, r) \neq (6, 1)$, the strictly second largest eigenvalue is $r(m-3)! = \mu_2(m, m-1; r)$ and can only be attained by $(m-1, 1), (2, 1^{m-2}), (m-2, 1^2), (3, 1^{m-3})$; hence $\mathbf{N}(m, m-1, r)$ holds. If moreover $3 \leq r \leq m-2$, it is attained exactly by $(m-1, 1)$ and $(2, 1^{m-2})$; hence $\mathbf{S}(m, m-1, r)$ holds.

Input 2.10 (Aldous property for $4 \leq k \leq n-2$ [3, Theorems 4.4, 4.6]). If $4 \leq k \leq n-2$ and $1 \leq r \leq k-1$, then $\mathbf{N}(n, k, r)$ holds.

Input 2.11 ([3, Lemmas 4.2, 4.3]). If $n \geq 8$ is even and $1 \leq r < (n-1)/2$, then $S(n, n-2, r)$ holds.

3. STRICT STEP

Lemma 3.1. *Let $2 \leq k \leq n-2$ and $1 \leq r \leq k-2$. If $S(n, k, r+1)$ and $N(n-1, k, r)$ hold, then $S(n, k, r)$ holds.*

Proof. Let $\lambda \notin \mathcal{D}_k(n) \cup \Theta_k(n)$ and split as in Lemma 2.6: $T_\lambda(n, k, r) = A + B$ with $A = \rho_\lambda(C_t^+)$ and $B = T_\lambda(n, k, r+1)$. By Lemma 2.4, λ has no trivial child, and if k is odd no sign child either; so every child ν satisfies $\nu \notin \mathcal{D}_k(n-1)$, and $N(n-1, k, r)$ gives $\lambda_{\max}(A) \leq \mu_2(n-1, k; r)$. By $S(n, k, r+1)$, $\lambda_{\max}(B) < \mu_2(n, k; r+1)$. By Lemma 2.3 and Input 2.8, $\Lambda_\lambda < \mu_2(n-1, k; r) + \mu_2(n, k; r+1) = \mu_2(n, k; r)$. $\square$

This is exactly the inductive step of [3, Theorems 4.4, 4.6], with the strict hypothesis on the second summand carried through.

4. THE STRICT BASE CASE $r = k-1$

4.1. Two zero-intersection lemmas. Throughout this subsection $N \geq 6$, $t \in [N]$, and $Y \subseteq [N]$ satisfies $t \in Y$ and $4 \leq |Y| \leq N-1$. Write $G_t = \text{Sym}([N] \setminus \{t\})$.

Lemma 4.1. *Let $W \subseteq S^{(N-2,2)}$ be the G_t -submodule isomorphic to the standard module $S^{(N-2,1)}$. Then W contains no nonzero vector fixed by $\text{Alt}(Y)$.*

Proof. Realise $S^{(N-2,2)}$ inside the two-subset permutation module $M = \mathbb{C}^{\binom{[N]}{2}}$ as the symmetric arrays $x = (x_{ij})$ with zero diagonal and zero row sums; the row-sum map $M \rightarrow \mathbb{C}^N$ is onto, and its kernel has dimension $N(N-3)/2$ and is the complement of the trivial and standard summands of $M \cong S^{(N)} \oplus S^{(N-1,1)} \oplus S^{(N-2,2)}$. For $u \in \mathbb{C}^{[N] \setminus \{t\}}$ with $\sum_i u_i = 0$ define

$$x_{ij}^u = u_i + u_j \quad (i \neq j, i, j \neq t), \quad x_{it}^u = x_{ti}^u = -(N-3)u_i.$$

Row $i \neq t$ sums to $(N-2)u_i - u_i - (N-3)u_i = 0$ and row t to $-(N-3)\sum_i u_i = 0$. The map $u \mapsto x^u$ is G_t -equivariant and injective (look at the entries x_{it}^u), so its image is a copy of the standard module of G_t inside $S^{(N-2,2)}$; by multiplicity-freeness it is W .

Reduction to $\text{Sym}(Y)$. Restricted to $\text{Sym}(Y)$, M is the sum of the permutation modules on the $\text{Sym}(Y)$ -orbits of two-subsets: the two-subsets of Y , the pairs $\{y, o\}$ with $y \in Y$ for each fixed $o \notin Y$, and the two-subsets of $[N] \setminus Y$. The point stabilisers are $\text{Sym}(Y \setminus \{a, b\}) \times \text{Sym}(\{a, b\})$, $\text{Sym}(Y \setminus \{y\})$ and $\text{Sym}(Y)$, each containing a transposition since $|Y| \geq 3$. By Frobenius reciprocity none of these permutation modules contains the sign character of $\text{Sym}(Y)$. Since the $\text{Alt}(Y)$ -fixed space of a $\text{Sym}(Y)$ -module is the sum of its trivial and sign isotypic parts, $M^{\text{Alt}(Y)} = M^{\text{Sym}(Y)}$, and hence $W^{\text{Alt}(Y)} = W^{\text{Sym}(Y)}$.

No $\text{Sym}(Y)$ -fixed vector. Suppose $x = x^u$ is fixed by $\text{Sym}(Y)$, i.e. $x_{\sigma i, \sigma j} = x_{ij}$ for $\sigma \in \text{Sym}(Y)$. Since $\text{Sym}(Y)$ is transitive on two-subsets of Y , all x_{ij} with $i, j \in Y$ equal a common value c' . For $a \in Y \setminus \{t\}$ this gives $-(N-3)u_a = c'$, so $u_a = c$ is independent of a ; choosing distinct $a, b \in Y \setminus \{t\}$ (possible since $|Y| \geq 3$) gives $2c = u_a + u_b = c' = -(N-3)c$, so $c = 0$. For $o \notin Y$, $\text{Sym}(Y)$ is transitive on $\{\{y, o\} : y \in Y\}$, so $x_{ao} = x_{to}$ for $a \in Y \setminus \{t\}$, i.e. $u_o = u_a + u_o = -(N-3)u_o$, whence $u_o = 0$. Thus $u = 0$. $\square$

Lemma 4.2. *Let $W \subseteq S^{(N-2,1,1)}$ be the G_t -submodule isomorphic to $S^{(N-2,1)}$. Then W contains no nonzero vector fixed by $\text{Alt}(Y)$.*

Proof. Let $V_N = \{z \in \mathbb{C}^N : \sum z_i = 0\}$, so $S^{(N-2,1,1)} \cong \Lambda^2 V_N$. Put $w = (N-1)e_t - \sum_{i \neq t} e_i$ and $V' = \{u \in V_N : u_t = 0\}$. Then $V_N = V' \oplus \mathbb{C}w$ as G_t -modules, with V' standard and w fixed, so $\Lambda^2 V_N = \Lambda^2 V' \oplus (V' \wedge w)$ and $W = V' \wedge w$ (again by multiplicity-freeness, as $\Lambda^2 V' \cong S^{(N-3,1,1)}$). Write $u \wedge w = \frac{1}{2} \sum_{i \neq j} c_{ij} e_i \wedge e_j$ with $c_{ji} = -c_{ij}$; then

$$c_{ij} = u_j - u_i \ (i, j \neq t), \quad c_{it} = (N-1)u_i \ (i \neq t).$$

Reduction to $\text{Sym}(Y)$. As a $\text{Sym}(Y)$ -module, $V_N \cong V_Y \oplus (N-|Y|) \cdot \mathbf{1}$, so $\Lambda^2 V_N \cong \Lambda^2 V_Y \oplus (N-|Y|)V_Y \oplus \binom{N-|Y|}{2} \mathbf{1}$. Neither $V_Y \cong S^{(|Y|-1,1)}$ nor $\Lambda^2 V_Y \cong S^{(|Y|-2,1,1)}$ is the sign module when $|Y| \geq 4$. As before, the $\text{Alt}(Y)$ - and $\text{Sym}(Y)$ -fixed spaces coincide.

No $\text{Sym}(Y)$ -fixed vector. If $\sigma(u \wedge w) = u \wedge w$ for all $\sigma \in \text{Sym}(Y)$, then $c_{\sigma i, \sigma j} = c_{ij}$. For $a \in Y \setminus \{t\}$, the transposition (at) gives $c_{ta} = c_{at} = -c_{ta}$, so $c_{at} = (N-1)u_a = 0$. For $o \notin Y$ and $a \in Y \setminus \{t\}$, the same transposition gives $c_{ao} = c_{to}$, i.e. $u_o - u_a = -(N-1)u_o$, so $u_o = 0$. Thus $u = 0$. $\square$

4.2. The strict base.

Proposition 4.3. *For every $k \geq 4$ and every $n \geq k+1$, $S(n, k, k-1)$ holds.*

Proof. Fix $k \geq 4$ and induct on n .

Base $n = k+1$. Here $r = k-1 = n-2$ and $n \geq 5$. If n is odd, $n/2 < n-2 < n-1$, and Input 2.9(a) gives $S(n, n-1, n-2)$. If n is even, then $n \geq 6$, $3 \leq n-2$, $(n, n-2) \neq (6, 1)$, and Input 2.9(b) gives $S(n, n-1, n-2)$.

Step. Let $n \geq k+2$ (so $n \geq 6$) and assume $S(n-1, k, k-1)$. Put $t = k$, $Y = [k]$, and split $C(n, k; k-1) = C_t \sqcup K_Y$ as in Lemma 2.6. Let $\lambda \notin \mathcal{D}_k(n) \cup \Theta_k(n)$ and write $T_\lambda(n, k, k-1) = A+B$, $A = \rho_\lambda(C_t^+)$, $B = \rho_\lambda(K_Y^+)$. By Input 2.7,

$$\mu_2(n, k; k-1) = (k-1)!(n-k) = \mu_2(n-1, k; k-1) + (k-1)!$$

The summand A . As in the proof of Lemma 3.1, no child ν of λ lies in $\mathcal{D}_k(n-1)$. By the induction hypothesis, $\Lambda_\nu(n-1, k, k-1) < \mu_2(n-1, k; k-1)$ unless $\nu \in \Theta_k(n-1)$, in which case equality holds. Hence $\lambda_{\max}(A) \leq \mu_2(n-1, k; k-1)$, and if equality holds then every top eigenvector of A lies in the sum of the isotypic components of the children belonging to $\Theta_k(n-1)$.

The summand B . By Lemma 2.5 ($|Y| = k \geq 4$), $\lambda_{\max}(B) \leq (k-1)!$, with top eigenvectors fixed by $\text{Sym}(Y)$ or $\text{Alt}(Y)$; in both cases they are fixed by $\text{Alt}(Y)$.

By Lemma 2.3, $\Lambda_\lambda \leq \mu_2(n, k; k-1)$, and equality would give a nonzero $\text{Alt}(Y)$ -fixed vector in a child component belonging to $\Theta_k(n-1)$. We show this is impossible.

If λ has no child in $\Theta_k(n-1)$, then $\lambda_{\max}(A) < \mu_2(n-1, k; k-1)$ and the inequality is strict. By Lemma 2.4, the remaining $\lambda \notin \Theta_k(n)$ are: $(n-2, 2)$ and $(n-2, 1, 1)$ (child $(n-2, 1)$), and, when k is odd, also $(2, 2, 1^{n-4})$ and $(3, 1^{n-3})$ (child $(2, 1^{n-3})$). Since $n \geq 6$, $(2, 1^{n-3})$ is not a child of $(n-2, 2)$ or $(n-2, 1, 1)$, so for these two shapes the vector would lie in the standard child, which is excluded by Lemmas 4.1 and 4.2 (applied with $N = n$, $|Y| = k \in [4, n-1]$). If k is odd, all elements of $C(n, k; k-1)$ are even, so T_λ is orthogonally similar to T_λ and $\Lambda_{(2,2,1^{n-4})} = \Lambda_{(n-2,2)}$, $\Lambda_{(3,1^{n-3})} = \Lambda_{(n-2,1,1)}$. Hence $\Lambda_\lambda < \mu_2(n, k; k-1)$ in all cases. $\square$

5. SMALL CYCLE LENGTH

For $k \in \{2, 3\}$ the published results give the Aldous property [1, 2] (see [3, Theorem 1.2 and the paragraph after it]); we prove the strict statement directly. Let $X_m = \sum_{i < m} (i \ m) \in \mathbb{C}[S_n]$ be the Jucys–Murphy elements. In Young’s orthogonal basis $\{v_T\}$ of S^λ , indexed by standard Young tableaux T of shape λ , $X_m v_T = c_T(m) v_T$, where $c_T(m)$ is the content (column minus row) of the box of T containing m ; and $s = (n-1 \ n)$ preserves each subspace $P_T = \text{span}\{v_T, v_{sT}\}$ (with v_{sT} omitted if sT is not standard), acting on a two-dimensional P_T as an involution with

eigenvalues 1 and -1 , and on a one-dimensional P_T as $+1$ (if $n-1, n$ lie in the same row of T) or -1 (same column). All sets below are conjugated by a permutation, which does not change spectra. For $\lambda \notin \{(n), (1^n), (n-1, 1), (2, 1^{n-2})\}$ we have $\lambda_1 \leq n-2$ and $\ell(\lambda) \leq n-2$, so every content lies in $[-(n-3), n-3]$.

Proposition 5.1. $S(n, 2, 1)$, $S(n, 3, 1)$ and $S(n, 3, 2)$ hold for every $n \geq 5$.

Proof. $k = 2, r = 1$. Conjugating $1 \leftrightarrow n$, $H^+ = X_n$, so the spectrum of T_λ is the set of contents of the removable corners of λ and $\Lambda_\lambda \leq \lambda_1 - 1$. Thus $\mu_2 = \Lambda_{(n-1, 1)} = n-2$, and for $\lambda \notin \{(n), (n-1, 1)\}$, $\Lambda_\lambda \leq n-3$.

$k = 3, r = 1$. With $X = \sum_{a \neq 1} (1 a)$, the products $(1 a)(1 b)$, $a \neq b$, run exactly once through the 3-cycles moving 1, so $H^+ = X^2 - (n-1)$. After conjugation, T_λ has eigenvalues $c^2 - (n-1)$, c ranging over contents of removable corners. The targets give $(n-2)^2 - (n-1) = \mu_2(n, 3; 1)$, while for $\lambda \notin \mathcal{D} \cup \Theta$ all contents satisfy $|c| \leq n-3$, so $\Lambda_\lambda \leq (n-3)^2 - (n-1) < \mu_2$.

$k = 3, r = 2$. One checks $(12)(1 a) = (1 a 2)$ and $(12)(2 a) = (1 2 a)$, so $H^+ = (12) \sum_{a \geq 3} ((1 a) + (2 a))$. Conjugating $\{1, 2\} \rightarrow \{n-1, n\}$ gives $H^+ = s(X_{n-1} + X_n - s) = s(X_{n-1} + X_n) - 1$. The element $Z = X_{n-1} + X_n$ acts on P_T as the scalar $\sigma_T = c_T(n-1) + c_T(n)$, so the eigenvalues of T_λ on P_T are $\sigma_T - 1$ (same row), $-\sigma_T - 1$ (same column), or $\pm\sigma_T - 1$ (otherwise). If $n-1, n$ occupy the last two boxes of row i , then $\sigma_T - 1 = 2(\lambda_i - i) - 2 \leq 2\lambda_1 - 4$. If they occupy the last two boxes of column j , then $-\sigma_T - 1 = 2(\lambda'_j - j) - 2 \leq 2\ell(\lambda) - 4$. Otherwise the two boxes lie in different rows and columns; if they are in rows $i_1 < i_2$, their contents are at most $\lambda_1 - 1$ and $\lambda_2 - 2$, so $\sigma_T \leq \lambda_1 + \lambda_2 - 3 \leq n-3$, and symmetrically $-\sigma_T \leq n-3$; so the eigenvalue is at most $n-4$. For $\lambda = (n-1, 1)$ the first case gives $2n-6 = \mu_2(n, 3; 2)$ (Input 2.7). For $\lambda \notin \mathcal{D} \cup \Theta$ the three bounds give $\Lambda_\lambda \leq \max(2n-8, n-4) < 2n-6$. $\square$

6. PROOF OF THEOREM 1.2

By Lemma 2.2 we must prove $S(n, k, r)$ for $n \geq 5$ and $1 \leq r < k \leq n-2$.

Case $k \leq 3$. Proposition 5.1.

Case $k \geq 4$ and $n-k \geq 3$. By Proposition 4.3, $S(n, k, k-1)$ holds. For $r = k-2, k-3, \dots, 1$ apply Lemma 3.1; its hypothesis $N(n-1, k, r)$ holds by Input 2.10, because $4 \leq k \leq (n-1)-2$.

Case $k \geq 4$ and $k = n-2$. Then $n \geq 6$, and Proposition 4.3 gives $S(n, n-2, n-3)$. Lemma 3.1 now needs $N(n-1, n-2, r)$, which concerns the layer $k = n-1$ of S_{n-1} .

- n odd, $n \geq 9$. Then $n-1 \geq 8$ is even and $N(n-1, n-2, r)$ holds for all r by Input 2.9(b); descend from $r = n-4$ to $r = 1$.
- $n = 7$. Descend to $r = 3, 2$ using $N(6, 5, 3)$, $N(6, 5, 2)$ (Input 2.9(b)). The case $r = 1$ is checked by computation (Section 7); it was also verified in Magma in [3, proof of Theorem 4.6].
- n even, $n \geq 8$. For $n/2 \leq r \leq n-4$ we have $(n-1)/2 < r < n-2$, so $N(n-1, n-2, r)$ holds by Input 2.9(a); descend from $r = n-4$ to $r = n/2$. The remaining values $1 \leq r \leq n/2 - 1$ satisfy $r < (n-1)/2$ and are covered by Input 2.11.
- $n = 6$. Here $k = 4$ and $r \in \{1, 2, 3\}$; $r = 3$ is Proposition 4.3, and $r = 1, 2$ are checked by computation (Section 7); all three were also verified in Magma in [3, proof of Theorem 4.4].

This covers every (n, k, r) with $n \geq 5$ and $1 \leq r < k \leq n-2$. $\square$

Remark 6.1. Only the base case needed a new idea. The published induction [3, Theorems 4.4, 4.6] already has the right shape; what it lacks is strictness at $r = k-1$, where [3, Theorem 1.2] gives only the Aldous property. Proposition 4.3 supplies this for every layer at once, because the equality analysis of Lemmas 4.1–4.2 does not depend on $n-k$. The layer $k = n-2$ is the only place where the argument touches the exceptions of [3, Theorem 1.3]; they are avoided by Input 2.11 and by the two small computations.

7. THE FINITE COMPUTATIONS

The proof uses computation only for $(n, k, r) \in \{(6, 4, 1), (6, 4, 2), (7, 5, 1)\}$. We recomputed all of $(6, 4, r)$, $r = 1, 2, 3$, and $(7, 5, r)$, $r = 1, \dots, 4$, in exact rational arithmetic:

- each irreducible representation was built in Young's seminormal form over $\mathbb{Q}$ and checked against the Coxeter relations and, on one element of every cycle type, against the Murnaghan–Nakayama rule;
- $T_\lambda = \sum_F \rho(\pi_F) D \rho(\pi_F)^{-1}$, where F runs over the $(n - k)$ -subsets of $[n] \setminus [r]$ (the possible fixed-point sets), π_F maps $\{k + 1, \dots, n\}$ onto F , and D is the diagonal action of the class sum of k -cycles of $\text{Sym}([k])$ in the seminormal basis;
- the characteristic polynomial of each T_λ was computed over $\mathbb{Q}$, and real roots in $[\mu_2, \infty)$ were counted by Sturm sequences.

In every case, each block $\lambda \notin \mathcal{D}_k(n) \cup \Theta_k(n)$ has no root in $[\mu_2, \infty)$, and each target block has μ_2 as its largest root.

(n, k, r)	degree	μ_2	attaining blocks	multiplicity of μ_2 in each
$(6, 4, 1)$	60	18	$(5, 1)$	4
$(6, 4, 2)$	36	16	$(5, 1)$	3
$(6, 4, 3)$	18	12	$(5, 1)$	2
$(7, 5, 1)$	360	84	$(6, 1), (2, 1^5)$	5
$(7, 5, 2)$	240	78	$(6, 1), (2, 1^5)$	4
$(7, 5, 3)$	144	66	$(6, 1), (2, 1^5)$	3
$(7, 5, 4)$	72	48	$(6, 1), (2, 1^5)$	2

As an independent sanity check (not used in the proof), a floating-point computation of all blocks confirmed $S(n, k, r)$ for every $5 \leq n \leq 10$ and $1 \leq r < k \leq n - 2$, and reproduced the exceptions of [3, Theorem 1.3] in the layer $k = n - 1$ for $5 \leq n \leq 9$.

PROVENANCE

The equality analysis in Section 4 generalises the argument of an earlier note, produced with the EULER agent system, that treated the single layer $n - k = 3$. The extension to all layers, the treatment of $k \leq 3$ and $k = n - 2$, and this write-up were prepared with AI assistance and have not yet been independently refereed. Novelty and priority have not been established by a literature search beyond the sources cited.

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